

Composing Piecewise and Nonlinear Functions

Answer Key

Precalculus Advanced Transformations and Composition

Quick Answers

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|--------------------------------|-------------------------------------|
| 1. 22 | 7. $\frac{1}{2}$ |
| 2. $4x^2 + 4x - 2$ | 8. 2 |
| 3. 9 | 9. -5, 5 |
| 4. $-\frac{5}{2}, \frac{3}{2}$ | 10. $x - 7$ |
| 5. 13 | 11. $-\sqrt{7}, \sqrt{7}$ |
| 6. $2x^2 - 5$ | 12. -3, $-\frac{3}{2}, \frac{1}{2}$ |
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Common wrong answers

1. If they got 3: composed in the wrong order and computed $g(f(2))$ instead of $f(g(2))$
 3. If they got 8: used the second piece $5 - u$ because the input $x = -2$ was compared with 2 instead of the inner output $g(-2)$
 5. If they got 46: composed in the wrong order and computed $f(g(3))$ instead of $g(f(3))$
 8. If they got 5: used the $u^2 + 1$ piece even though the inner value $f(1)$ is negative
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1. $(f \circ g)(2)$.

Step 1. The inner function acts first: $g(2) = 2(2) + 1 = 5$.

Step 2. Feed that output into f : $f(5) = 5^2 - 3 = 25 - 3$.

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Common wrong answer: 3 — the order was reversed, computing $g(f(2)) = g(1) = 3$.

2. A single rule for $f(g(x))$.

Step 1. Substitute the whole inner rule into f 's variable slot: $f(g(x)) = (2x + 1)^2 - 3$.

Step 2. Expand the square before subtracting: $(2x + 1)^2 = 4x^2 + 4x + 1$, so $f(g(x)) = 4x^2 + 4x + 1 - 3$.

$4x^2 + 4x - 2$

Check: at $x = 2$ this gives $16 + 8 - 2 = 22$, matching problem 1.

3. $p(g(-2))$ with a piecewise outer function.

(a) $g(-2) = 2(-2) + 1 = -3$.

(b) The condition on p tests *its own input*, which is now -3 . Since $-3 < 2$, the first piece x^2 applies. (The original input -2 is irrelevant to that test.)

(c) $p(-3) = (-3)^2$.

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Common wrong answer: 8 — the second piece $5 - u$ was used, because the test was applied to $x = -2$ instead of to $g(-2) = -3$.

4. Solve $(f \circ g)(x) = 13$.

Step 1. Use the rule from problem 2: $4x^2 + 4x - 2 = 13$, so $4x^2 + 4x - 15 = 0$.

Step 2. Factor: $4x^2 + 4x - 15 = (2x + 5)(2x - 3)$, so $2x + 5 = 0$ or $2x - 3 = 0$.

$$x = -\frac{5}{2} \quad \text{or} \quad x = \frac{3}{2}$$

Check: $g(-\frac{5}{2}) = -4$ and $f(-4) = 13$; $g(\frac{3}{2}) = 4$ and $f(4) = 13$.

5. $(g \circ f)(3)$.

Step 1. Inner first: $f(3) = 3^2 - 3 = 6$.

Step 2. Then $g(6) = 2(6) + 1$.

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What the comparison shows: problem 1 gave 22 from the same two functions in the other order. Composition is **not** commutative — the order is part of the operation.

Common wrong answer: 46, which is $f(g(3))$.

6. A single rule for $g(f(x))$.

Step 1. Substitute f 's rule into g : $g(f(x)) = 2(x^2 - 3) + 1$.

Step 2. Distribute and collect: $2x^2 - 6 + 1$.

$$2x^2 - 5$$

Compare with problem 2: $f(g(x)) = 4x^2 + 4x - 2$ has a linear term and a different leading coefficient. Two different functions, so $f \circ g \neq g \circ f$ in general.

7. Where $(p \circ g)$ switches pieces.

(a) The pieces of p are separated by the value 2 of its input, and its input here is $g(x) = 2x + 1$.

So solve $2x + 1 = 2$, giving $2x = 1$.

$$x = \frac{1}{2}$$

(b) For $x < \frac{1}{2}$ the inner value $2x + 1$ is below 2, so the first piece applies; for $x \geq \frac{1}{2}$ the second piece applies:

$$(p \circ g)(x) = \begin{cases} (2x + 1)^2, & x < \frac{1}{2} \\ 5 - (2x + 1) = 4 - 2x, & x \geq \frac{1}{2} \end{cases}$$

Note the shift: the break in p sits at 2, but the break in $p \circ g$ sits at $\frac{1}{2}$ — the inner function moves it.

8. $(q \circ f)(1)$.

(a) $f(1) = 1^2 - 3 = -2$.

(b) q tests its own input against 0. That input is -2 , which is negative, so the first piece $-u$ applies.

(c) $q(-2) = -(-2)$.

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Common wrong answer: 5 — the $u^2 + 1$ piece was used because $x = 1$ is positive. The test never looks at x ; it looks at what reaches q .

9. Solve $(g \circ f)(x) = 45$.

Step 1. From problem 6, $2x^2 - 5 = 45$, so $2x^2 = 50$ and $x^2 = 25$.

Step 2. Take both roots — $x^2 = 25$ has two real solutions:

$$\boxed{x = -5 \quad \text{or} \quad x = 5}$$

Why two: the inner function f is quadratic, so it takes the value 25 at two inputs. A composition of linear functions is linear and can only cross a given output once.

Check: $f(\pm 5) = 22$ and $g(22) = 45$.

10. A single rule for $f(h(x))$, and its domain.

(a) $f(h(x)) = (\sqrt{x-4})^2 - 3$. Squaring undoes the square root on the root's own domain, leaving $(x-4) - 3$. $\boxed{x-7}$

(b) Domain: h requires $x-4 \geq 0$, so $x \geq 4$, i.e. $[4, \infty)$. The simplified rule $x-7$ is defined for every real x , so the simplification *threw away the inner function's domain restriction*. The composition is $x-7$ only on $[4, \infty)$ — a line with its left half deleted, not a line.

Rule to remember: always take the domain from the composition *before* simplifying.

11. Domain of $(h \circ f)(x) = \sqrt{x^2 - 7}$.

(a) The radicand must be non-negative, so the boundary is where it is zero: $x^2 - 7 = 0$, giving $x^2 = 7$.

$$\boxed{x = -\sqrt{7} \quad \text{or} \quad x = \sqrt{7}}$$

(b) $x^2 - 7 \geq 0$ holds outside those boundaries, so the domain is

$$\left(-\infty, -\sqrt{7}\right] \cup \left[\sqrt{7}, \infty\right).$$

It comes in two pieces because the inner function is an upward parabola: it dips *below* the required level between its two crossing points, and rises above it on both outer arms. (Decimal check: $\sqrt{7} \approx 2.65$, and $x = 2$ gives a radicand of -3 .)

12. Solve $(q \circ g)(x) = 5$ with a piecewise outer function.

Step 1. q receives $u = 2x + 1$, so each piece must be solved separately *and then tested* against the condition that selects it.

(a) **First piece** ($u < 0$, rule $-u$): $-(2x+1) = 5$ gives $-2x-1 = 5$, so $-2x = 6$. $\boxed{x = -3}$

Test: $u = 2(-3) + 1 = -5$, which is negative, so this piece really does apply. **Accepted.**

(b) **Second piece** ($u \geq 0$, rule $u^2 + 1$): $(2x+1)^2 + 1 = 5$ gives $(2x+1)^2 = 4$, so $2x+1 = 2$ or $2x+1 = -2$:

$$\boxed{x = \frac{1}{2} \quad \text{or} \quad x = -\frac{3}{2}}$$

Test each: $x = \frac{1}{2}$ gives $u = 2 \geq 0$ — **accepted**. $x = -\frac{3}{2}$ gives $u = -2 < 0$ — **rejected**, because that value of x is not in the region where the $u^2 + 1$ piece is in force.

(c) Complete solution set: $x = -3$ and $x = \frac{1}{2}$.

Why one candidate had to go: solving a piece's equation only asks “what makes this formula equal 5?”. It does not ask “is this formula the one q actually uses there?”. The second question is the domain test, and skipping it is the standard error on piecewise equations.